Code No: 118BA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) With suitable examples, explain the security problems in databases and threats to database security.
b) Differentiate between mandatory and discretionary access controls. [7+8]

2.a) Does access matrix model handle Trojan horses? Justify your answer.
b) Predicates on authorizations are included in Acten model. What are the benefits of this inclusion? Explain with example. [7+8]

3. Explain the Bell and LaPadula's model. [15]

4.a) What is a multilevel relation? Discuss a security model to secure them.
b) Make a comparison of different user identification/authentication system. [8+7]

5.a) Discuss the advantage of a security kernel in an operating system and properties of security kernel.
b) What is a covert channel? How was it analyzed during VAX security kernel development? [7+8]

6.a) Contrast operating system security with database security.
b) Explain types of attacks. [8+7]

7. Discuss the main elements and concepts of object-oriented databases and a model for the protection of these databases. [15]

8.a) What are condition predicates? Compare them with transition predicate.
b) How information flow is controlled in Jajodia and Kogan model for object oriented systems. [7+8]

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